

SREEHITHA REDDY NARAYANA

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Education

University of Massachusetts Amherst

Aug 2024 – May 2026

Master of Science in Computer Science — GPA: 3.81/4.0

Amherst, MA

- **Coursework:** Machine Learning, Systems for Data Science, Advanced NLP, Database Design, Distributed Systems

Indian Institute of Technology, Hyderabad

Jul 2019 – May 2023

Bachelor of Technology in Computer Science

Hyderabad, India

- Awarded Panasonic Ratti Chhatr Scholarship for academic excellence
- **Coursework:** Data Structures, Algorithms, Probability, Operating Systems, Computer Networking, Software Engineering

Experience

Dolby Laboratories

Jan 2026 – May 2026

Research Extern — [\[arXiv\]](#)

Remote, USA

- Developed an **LLM-based multimodal music recommendation** system for session-based next-song prediction, combining sequential behavior modeling with content-aware item representations across SASRec, BERT4Rec, and GRU4Rec backbones
- Built a large-scale **LastFM-1K** enrichment pipeline over **4.2M+ listening events**, 814 users, and 295K+ tracks, generating a Top-50K song catalog with **audio, lyric, metadata, and completion-ratio features**.
- Integrated pretrained audio and lyric encoders with **GPT-5** generated 58-attribute musicological metadata, capturing acoustic, lyrical, semantic, and engagement-aware signals for **content-aware** ranking.
- Fine-tuned and evaluated LLM recommenders using **LoRA** across LLaMA, Qwen, and multiple embedding fusion strategies, on **SLURM** with **CUDA-enabled GPUs**, using Recall, NDCG.

Jio Platforms Ltd

Jul 2023 – Aug 2024

Software Engineer

Bangalore, India

- Engineered high-performance C++ backend services for large-scale **5G core systems**, improving reliability by 15%.
- Optimized data pipelines and performed deep network traffic analysis using **Wireshark**, identifying bottlenecks and reducing issue resolution time by 30% while improving throughput.
- Designed data flow across 5G components (**SMF/SBC**), ensuring **low-latency, fault-tolerant** communication.
- Delivered production-ready features in Agile sprints, working with cross-functional teams to strengthen system stability through code reviews and testing.

Technical Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript

Machine Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, SMOTE

LLMs & Generative AI: Hugging Face Transformers, LoRA Fine-Tuning, Prompt Engineering, RAG, FAISS

Data Engineering & Big Data: PySpark, Apache Kafka, PostgreSQL, MySQL, SQLite, NoSQL, Cassandra

MLOps & Cloud: AWS, Azure, Git, Linux, Docker, SLURM, CUDA

Software Engineering: REST APIs, Node.js, Express.js, React.js, JUnit, Maven, Gradle

AI-Assisted Development: GitHub Copilot, Cursor, Claude Code

Projects

Medical SOAP Note Generation with knowledge Graph | *PyTorch, Transformers, NLP*— [github](#) Sep 2025 – Dec 2025

- Built an LLM-based pipeline using **T5-Large** to generate structured medical SOAP notes from doctor–patient dialogues.
- Fine-tuned the model using **LoRA** on domain specific dialogues to enhance model efficiency, and performance.
- Applied **prompt engineering** & KG augmented prompting to **reduce hallucinations** and improve clinical grounding.
- Reduced hallucination rate from 55% to 21% & improved BERTScore from 0.71 to 0.87 compared to text-only baselines.
- Evaluated models using metrics like ROUGE, BERTScore & LLM-as-judge for fluency and coherence.

Credit Card Fraud Detection System | *PySpark, Random Forest, AWS, Apache Kafka*— [github](#) Aug 2024 – Dec 2024

- Built a real-time fraud detection pipeline using **PySpark** and **Apache Kafka**, processing 10,000+ transactions/sec.
- Developed python services with SQL-based feature extraction & validation, achieving 98% precision on imbalanced data
- Deployed, monitored, & scaled the system on **AWS EC2**, ensuring reliability, fault tolerance and production readiness.

Distributed Middleware over Cassandra | *Java, Apache Cassandra, RPC*— [github](#) Sep 2025 – Dec 2025

- Developed a distributed middleware over Cassandra ensuring totally ordered writes & strong consistency across nodes.
- Designed a non-blocking **RPC** framework to coordinate requests and maintain synchronized updates among replicas.
- Implemented replication and ordering protocols to achieve a **consistent** global state before client acknowledgment.
- Supported **fault-tolerant** and **concurrent operations** through asynchronous request handling and coordinated execution.

UniTrade - UMass Marketplace | *Next.js, TypeScript, Node.js, Express, Prisma, PostgreSQL* – [github](#) Sep 2025 – Dec 2025

- Built a campus-exclusive **full-stack marketplace** serving university students, implementing **JWT-based** authentication and email verification, enabling UMass users to seamlessly buy, sell, and auction items.
- Designed scalable RESTful APIs and a normalized PostgreSQL schema for bidding, messaging, and admin moderation.
- Executed a real-time auction engine using **WebSockets** with live bidding, instant price updates, notifications, and **AI-powered** semantic search using **Google Gemini** to enhance product discovery.

High-Performance Database Indexing System | *Java, B+ Trees, Buffer Management* – [github](#) Jan 2025 – May 2025

- Built a disk-based **B+ Tree indexing** system for 100K+ records with optimized insert, search, and range query support.
- Built a **buffer manager** with **LRU eviction** and configurable memory limits (1MB–1GB) to reduce disk access latency.
- Developed a query execution engine using pipelined relational operators (Scan, Selection, Projection, BNL Join) for efficient multi-stage join processing on large datasets.

Scalable Distributed Vector Search System | *FAISS, HNSW, Python* – [github](#) May 2025 – Aug 2025

- Designed a distributed vector search engine for scalable **ANN** retrieval on 10M+ vectors using **HNSW** indexing.
- Working on coarse quantization and selective multi-host querying to balance recall and latency in **distributed** settings.
- Benchmarked recall, latency, and throughput using self-developed evaluation tools across single and multi-host setups.
- Integrated **FAISS** with a Python coordination layer for parallel query dispatch, result merging, and distributed evaluation.

Publication Management System | *Java, DSpace, API Integration* Jan 2023 – May 2023

- Built a Java-based **web application** managing 5,000+ research publications using the DSpace digital repository for reliable storage, indexing, and retrieval of academic content.
- Optimized backend services & database interactions, improving processing performance and reducing latency by 40%.
- Implemented **REST API** integrations to automate metadata ingestion & synchronization ensuring data consistency.

Leadership & Activities

- Impacted 500+ underprivileged students through educational outreach, mentorship, and community development initiatives
- Led a university-wide technical competition, driving a 30% increase in participation through strategic planning and outreach